From Maxwell's Rod to Unruh's Thermometer

A Post-Relativistic Basis for Physical Dimensions

Lee McCulloch-James

Department of Natural Sciences, Dwight School London

Abstract

Dimensional analysis rests on three base dimensions, mass, length and time, that James Clerk Maxwell fixed in 1873. The method that uses them was formalised by Buckingham and Bridgman decades later, after special relativity had shown that length and time are not quantities all observers agree on. The base was never revisited. This essay asks what a dimensional basis chosen after Einstein would look like. It keeps only what relativity leaves invariant: the spacetime interval, which fixes the speed of light and merges length with time, and proper acceleration, which survives as a covariant scalar where velocity does not. Adding temperature, measured as energy, closes a two-quantity basis in which mass is energy and length-time is inverse acceleration. The Unruh effect supplies the constant relating acceleration to thermal energy, and a short dimensional argument shows that constant must be Planck's; the same effect explains why the two quantities are canonically paired without being made linearly dependent. The exercise reframes Boltzmann's constant as a unit convention and locates dimensional analysis, usually taken for bookkeeping, as a question about which quantities relativity permits us to call fundamental.

1 A basis older than relativity

When a student first balances the dimensions of an equation, checking that force really does come out as mass times length over time squared, they are using a scheme that James Clerk Maxwell set down in the opening chapter of his *Treatise on Electricity and Magnetism* in 1873 [1]. Maxwell argued that all physical quantities can be expressed in terms of three fundamentals, which he took to be mass, length and time. A rod fixes the length, a clock the time, a prototype the mass, and every other quantity follows as a product of powers of these three. The scheme has served for a century and a half, and it works.

There is a curiosity in its history that is rarely remarked. Maxwell wrote in 1873, before relativity, so his treatment of length and time as absolute standards, the same for every observer, was the only treatment available to him. But the mathematical machinery that turned his scheme into the modern tool, Buckingham's π theorem of 1914 [2] and Bridgman's lectures of 1922 [3], was written after Einstein. By the time dimensional analysis was formalised, special relativity had been in the literature for nine and then seventeen years, and it had shown that a length is contracted and a time dilated by the Lorentz factor $\gamma = (1 - v^2/c^2)^{-1/2}$, so that neither the metre nor the second is a quantity on which observers in relative motion agree.

Historical observation

Maxwell chose M, L, T in 1873. Buckingham (1914) and Bridgman (1922) formalised the method after Einstein, and kept the same base. The choice is pre-relativistic; the formalism that fixed it is not.

Bridgman, who thought harder than most about what it means to define a physical quantity by the operations used to measure it, might have been expected to notice that two of his three base quantities had just lost their observer-independence. He did not revisit them, and neither has anyone since. The γ that governs the disagreement between observers never enters a dimensional analysis; we write $[L] = L$ and $[T] = T$ as though 1905 carried no lesson for the choice of base. The results stay correct, because dimensional analysis tracks the structure of a unit and not its behaviour under a change of frame. Yet the base has been inherited from quantities that relativity marks as belonging to the observer rather than to the world.

The question this essay pursues is a simple one. If we were to choose a dimensional basis today, keeping only what relativity certifies as invariant, what would it contain? The answer turns out to be shorter than Maxwell's list, and to place at its centre a quantity Maxwell never considered fundamental.

2 What relativity leaves standing

Special relativity does not leave the measurer empty-handed. It removes the absoluteness of length and time, but in the same stroke it supplies quantities that all inertial observers agree on, and these are the natural candidates for a modern base.

The first is the spacetime interval. Where two observers disagree about a spatial separation dx and a temporal one dt, they agree on the combination

$$ds^2 = c^2\, dt^2 - dx^2,$$

which is the same in every inertial frame. Taking the interval as the fundamental unit block, in place of length and time as separate standards, is the dimensional expression of what relativity actually asserts. In practice this is the familiar step of setting $c \equiv 1$: because the speed of light is the one velocity every observer shares, fixing it declares that space and time are measured in a single unit, related along the light-cone. Length and time cease to be two dimensions and become one, which we may write X, with $[L] = [T] = X$.

Key consequence

Fixing $c \equiv 1$ is not a computational shortcut. It is the statement that the interval, not the rod or the clock, is the invariant, and it reduces the mechanical base from three dimensions to two.

The second surviving invariant is more surprising, and it is the pivot of the whole construction. Velocity is exactly the quantity the Lorentz transformation alters; there is no absolute velocity, which is the content of the principle of relativity itself. Position and duration are frame-dependent for the same reason. But *proper acceleration*, the acceleration an observer reads on an accelerometer carried in their own frame, is a covariant scalar. It is the magnitude of the four-acceleration, and every observer computes the same value for it. In a theory that has stripped away absolute velocity, absolute length and absolute time, absolute acceleration remains.

That acceleration should outrank velocity as a fundamental quantity is counter-intuitive, because our pre-relativistic instincts treat velocity as the simpler notion. Relativity inverts the ranking. Uniform velocity is undetectable, indeed meaningless without a reference frame, whereas acceleration is felt, measured, and agreed upon. A basis chosen for invariance rather than familiarity puts acceleration in and leaves velocity out.

3 Closing the basis with temperature

Two quantities remain to be settled: the source of mass, and the treatment of temperature. They turn out to be the same question.

Merging length and time leaves the mechanical world two-dimensional, mass and the combined X. Acceleration supplies one of these, since with $c = 1$ its dimension is that of inverse length, $[A] = X^{-1}$. Mass has still to come from somewhere, and no combination of acceleration and the interval will produce it, because neither carries mass. The missing ingredient arrives with temperature, provided temperature is understood correctly.

A temperature is, physically, an energy per degree of freedom; the kelvin is a historical accident of having measured heat before recognising it as molecular motion. Measured in its natural currency, temperature is an energy, with dimension ML^2T^{-2}. Written this way it carries a factor of mass, and that is the factor the basis was missing. Taking the thermal quantity Θ with the dimension of energy, and working with $c = 1$, the two base quantities (A, Θ) close the mechanical world. Acceleration has dimension X^{-1} and thermal energy has dimension M, so

$$\mathbf{D} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \det \mathbf{D} = 1, \qquad \begin{aligned} M &= \Theta, \\ X &= A^{-1}. \end{aligned}$$

Mass is energy, and length-time is inverse acceleration. The entire mechanical world is generated by two invariant quantities once the separate standard of length, the last of Maxwell's three absolutes, is given up.

Key consequence

In the post-relativistic basis, mass is energy ($M = \Theta$) and length-time is inverse acceleration ($X = A^{-1}$). Two invariants replace Maxwell's three.

This treatment of temperature has an immediate corollary. If the thermal quantity is an energy from the start, then Boltzmann's constant, whose job is to convert kelvin into joules, has nothing to convert. It does not appear in the basis. This is the natural endpoint of a view now enshrined in the SI, which since 2019 has defined the kelvin by fixing the numerical value of k_B [5]: a defined conversion factor rather than a measured constant. Measuring temperature directly in energy completes the move, and k_B is absent rather than set to one.

4 The effect of the Unruh effect

The two base quantities are related by a law of physics, and the law carries a constant. An observer with proper acceleration a in the Minkowski vacuum registers a temperature given by the Unruh effect [4],

$$T(\hbar, k_B, a) = \frac{\hbar\, a}{2\pi\, k_B\, c}. \tag{1}$$

Both Planck's constant and Boltzmann's constant appear, and a reader tracking the basis might protest that neither was among the two quantities we began with. The protest is met by dimensional analysis itself.

Measure temperature as energy, absorbing k_B, and the relation reads $\Theta = \hbar a/2\pi c$. In the $c = 1$ basis, acceleration is an inverse length and thermal energy is a mass. To connect an inverse length to a mass requires a constant of dimension ML, and with c fixed the only such constant available is $\hbar/c$:

$$[\hbar/c] = \frac{ML^2T^{-1}}{LT^{-1}} = ML.$$

Planck's constant is not an intruder in Eq. (1); it is the mandatory bridge between the two base quantities. Any law relating proper acceleration to thermal energy must carry exactly one factor of $\hbar$, whatever its detailed physics, because the dimensions leave no alternative. The Unruh effect fixes the pure number in front, $1/2\pi$, and nothing else about the relation is free.

Key consequence

The appearance of $\hbar$ is forced by the dimensions. Acceleration is an inverse length, energy is a mass, and only $\hbar/c$ converts one to the other. Physics supplies the coefficient $1/2\pi$; the constant itself is mandatory.

5 A pairing, not a dependence

A question now presses. If a law of physics relates the two base quantities, are they really independent? In linear algebra, two vectors related by $\mathbf{v}_1 = \lambda \mathbf{v}_2$ are dependent, and cannot both belong to a basis. Does Unruh collapse (A, Θ) in the same way?

Dimensional analysis has always answered such questions differently from linear algebra, and the difference is instructive. The metre and the second are treated as independent base dimensions throughout physics, even though the speed of light provides a universal conversion between them. Kelvin and joule were held independent for a century despite the existence of k_B. A conversion constant preserves the independence of two units rather than destroying it, because it is an empirical constant and not a definition. The metre is related to the second by a measured c; it is not defined as a second times a number.

The decisive question is therefore whether the Unruh relation is a universal conversion of that kind. It is not. The relation $L = cT$ holds for every length and every time, without exception, which is what allows c to serve as a conversion while metre and second remain independent. Equation (1) holds only for uniformly accelerated observers in the Minkowski vacuum. A hot gas at rest has temperature and no acceleration; an accelerating spacecraft has acceleration and a wholly negligible Unruh temperature. Across the range of physical situations the correspondence between acceleration and temperature is many-to-many, an identity confined to one special class of states.

Acceleration and thermal energy are independent invariant quantities that become canonically identified through Unruh effect.

The right description of the Unruh relation is that it identifies a distinguished correspondence between two independent quantities, in the way that $E = hf$ identifies a distinguished correspondence between energy and frequency. No one concludes from $E = hf$ that energy and frequency are the same dimension; Planck's constant fixes a canonical pairing between them, holding wherever quantum mechanics applies. In just the same way, $k_BT = \hbar a/2\pi c$ fixes a canonical pairing between thermal energy and proper acceleration, holding on the submanifold of accelerated vacuum states. The two quantities remain independent, and the dimensional closure of the previous section stands. The basis is not constrained by a law of nature, any more than the SI metre-second basis is constrained by the existence of light.

6 The cold vacuum basis

One feature of the construction deserves to be drawn out, because it is the clearest expression of what the whole scheme is built on. The temperature in the Unruh relation is observer-relative. For a uniformly accelerated observer it takes the value in Eq. (1); for an inertial observer in the same vacuum it is exactly zero. The vacuum is cold to anyone in free fall and warm only in proportion to proper acceleration.

The single thermodynamic quantity in the basis therefore vanishes for inertial observers and, for accelerated ones, measures precisely the single kinematic invariant the theory retains. A basis assembled from relativity's invariants ends by tying its one thermal quantity to its one kinematic quantity, through the vacuum itself. Where Maxwell's scheme rested on three absolutes, this one rests on two invariants, the interval and the acceleration, with energy standing in for mass and the Unruh relation binding acceleration to temperature through a constant the dimensions demand.

Key consequence

The one temperature in the basis is zero for inertial observers and proportional to acceleration for the rest. The thermal and kinematic quantities are tied together through the vacuum.

7 A basis for line bundles

The reconstruction leaves dimensional analysis looking less like bookkeeping than it usually appears. The choice of base dimensions is a physical question about which quantities the world permits us to regard as fundamental, and the answer changed in 1905 without the practice of dimensional analysis registering the change.

A post-relativistic basis keeps the interval, which fixes c and merges length with time; proper acceleration, the kinematic invariant that outlives velocity; and thermal energy, which supplies mass and dispenses with Boltzmann's constant. Planck's constant enters as the forced bridge between acceleration and energy, with the Unruh effect setting its coefficient and explaining why the pairing is canonical without being an identity. Two invariant quantities do the work of Maxwell's three.

The value of the exercise is in the question it forces: a basis is a claim about what is fundamental, and relativity has views about that which a century of dimensional analysis has not consulted. This essay develops one strand of a longer programme that treats the choice of dimensional basis as a subject in its own right, alongside the mapping of mass and entropy scales onto a two-parameter family [6], the cosmological consequences of a Hubble-adapted basis [7], and the number-theoretic structure of dimensionless combinations [8].

The gauge-theoretic reading of dimensional analysis that underlies this view, with the scale group as gauge group and a unit system as a chosen gauge, is developed by Muntz [9]; the question this program adds to it is whether the choice of gauge, the basis itself, carries physical content beyond the algebra. Muntz begins with geometry and recovers dimensional analysis whereas here in the final section we approach from the opposite direction: beginning with Buckingham elimination and asking whether the familiar linear algebra is already hinting at line bundles, quotient spaces and gauge choices

8 Inverse temperature

The preceding discussion treated temperature as an independent dimensional direction and asked how it becomes identified with quantities in the mechanical sector. Before pursuing that question geometrically, however, it is useful to reconsider which thermal variable is primitive. In equilibrium statistical mechanics, the variable that enters the formal structure most directly is not T, but the inverse thermal parameter

$$\beta = \frac{1}{k_\mathrm{B}T}.$$

It is β that appears in the Boltzmann weight,

$$\exp(-\beta E),$$

and in the canonical partition function,

$$Z(\beta) = \sum_i \exp(-\beta E_i).$$

The combination βE is dimensionless. Geometrically, this suggests that energy and inverse temperature occupy dual dimensional slots: energy is a section of an energy line bundle L_E, while β is naturally a section of its dual L_E^*. Their pairing

$$L_E^* \otimes L_E \longrightarrow \mathbb{R}, \qquad (\beta, E) \longmapsto \beta E,$$

is canonical and requires no prior identification of temperature with energy.

This dual interpretation is also suggested by thermodynamics itself. If entropy is measured as a dimensionless state-counting quantity, then

$$\beta = \frac{\partial S}{\partial E}$$

is the covector conjugate to energy. More generally, the differential of the entropy takes the form

$$dS = \beta\, dE + \beta p\, dV - \beta\mu\, dN + \cdots,$$

so the intensive variables appear naturally as coefficients of differentials of the extensive variables. They therefore belong first to a cotangent description of thermodynamic state space.

On this view, k_B does more than replace kelvin by joules. It relates the operational temperature coordinate supplied by thermometry to the inverse-energy covector that enters the statistical and Legendre structure:

$$T \xrightarrow{k_\mathrm{B}} \beta^{-1}.$$

The distinction matters because T is obtained operationally through thermal equilibrium, level populations and fluctuation statistics, whereas energy is measured within the mechanical sector. The map between them is therefore a physically supplied identification between differently organised quantities, rather than the direct comparison of two measurements of the same operational kind.

This also clarifies the rôle of particular relations such as equipartition and the Unruh effect. Equipartition gives

$$\langle E \rangle = \frac{1}{2} k_\mathrm{B} T$$

per quadratic degree of freedom, while the Unruh relation gives

$$\beta = \frac{1}{k_\mathrm{B} T} = \frac{2\pi c}{\hbar a}.$$

Each supplies a physically meaningful identification of the thermal direction with a mechanical quantity, but only within a restricted regime. Equipartition applies on the equilibrium submanifold of systems with quadratic Hamiltonians; the Unruh relation applies to uniformly accelerated observers in the quantum vacuum. These are therefore regime-dependent trivialisations of the thermal line, rather than a global identification of temperature with either energy or acceleration.

8.1 Towards symplectic thermodynamics

Once inverse temperature is recognised as a covector conjugate to energy, the line-bundle picture opens naturally onto the geometric formulation of thermodynamics. The thermodynamic phase space is an odd-dimensional contact manifold equipped with a contact one-form. In the entropy representation one may write schematically

$$\eta = dS - \beta\, dE - \beta p\, dV + \beta\mu\, dN - \cdots.$$

Equilibrium states lie on Legendre submanifolds for which

$$\eta = 0.$$

The first law and the equations of state are thereby encoded as geometric constraints rather than appended relations.

The passage from the contact manifold to its symplectisation introduces an even-dimensional space with a symplectic two-form derived from the contact structure. At a later stage one may add further geometric data, such as the Fisher, Weinhold or Ruppeiner metrics, in order to define lengths, angles and curvature on thermodynamic state space. No particular metric need be chosen for the present argument. The essential point is the hierarchy

$$\begin{aligned}\text{dimensional line bundles} &\to \text{dual energy–thermal pairing}\\ &\longrightarrow \text{contact structure}\\ &\longrightarrow \text{symplectic and metric thermodynamics.}\end{aligned}$$

The line-bundle language therefore supplies a natural introductory route into geometric thermodynamics. Energy and inverse temperature first appear as sections of dual lines whose canonical pairing produces the dimensionless exponent βE. The contact form then assembles these pairings into the local structure of equilibrium thermodynamics. Symplectic and metric constructions arise as subsequent enlargements of the same geometric programme.

For readers accustomed to dimensional quantities as sections of line bundles, the cotangent pairing of energy and inverse temperature provides a natural passage into contact and symplectic thermodynamics. Geometric thermodynamics is therefore less an imported formalism than the next geometric enlargement of the same idea.

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